

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSA0307 - Data Structures

Title on

Dijkstra's Algorithm for Delivery Route Optimization

By

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Objective

- Model cities, warehouses, and delivery points as nodes with weighted edges (distance, time, fuel cost) in a graph.
- Apply Dijkstra's Algorithm to compute the shortest path from a central warehouse to multiple delivery points.
- Provide a GUI with map-based input and animated display of the computed routes.
- Estimate delivery time and fuel cost, and benchmark results against brute-force and greedy approaches.
- Support real-time rerouting using dynamic traffic and weather data.



Literature Review

S.No	Title of the paper	Published Year	Components /Techniques used
1	Shortest Path Optimization for Last-Mile Delivery using Dijkstra's Algorithm	2023	Weighted graph modeling, Dijkstra's algorithm, priority queue
2	Comparative Study of Dijkstra, Bellman-Ford and A* for Logistics Routing	2024	Graph algorithms, benchmarking, complexity analysis
3	GIS-Based Route Optimization System for Fleet and Delivery Management	2024	GIS mapping, GPS integration, path visualization
4	Real-Time Traffic-Aware Route Rescheduling for E-Commerce Deliveries	2025	Dynamic graph weights, live traffic API, adaptive rerouting



Problem Statement

- Manual or heuristic route planning rarely identifies the true shortest or fastest path across a large delivery network.
- Sub-optimal routes increase fuel consumption, delivery time, and overall operating cost for logistics companies.
- Routes are fixed in advance and cannot respond to changing traffic, weather, or road closures once dispatched.
- There is no systematic comparison between brute-force, greedy, and optimal (Dijkstra) routing strategies.

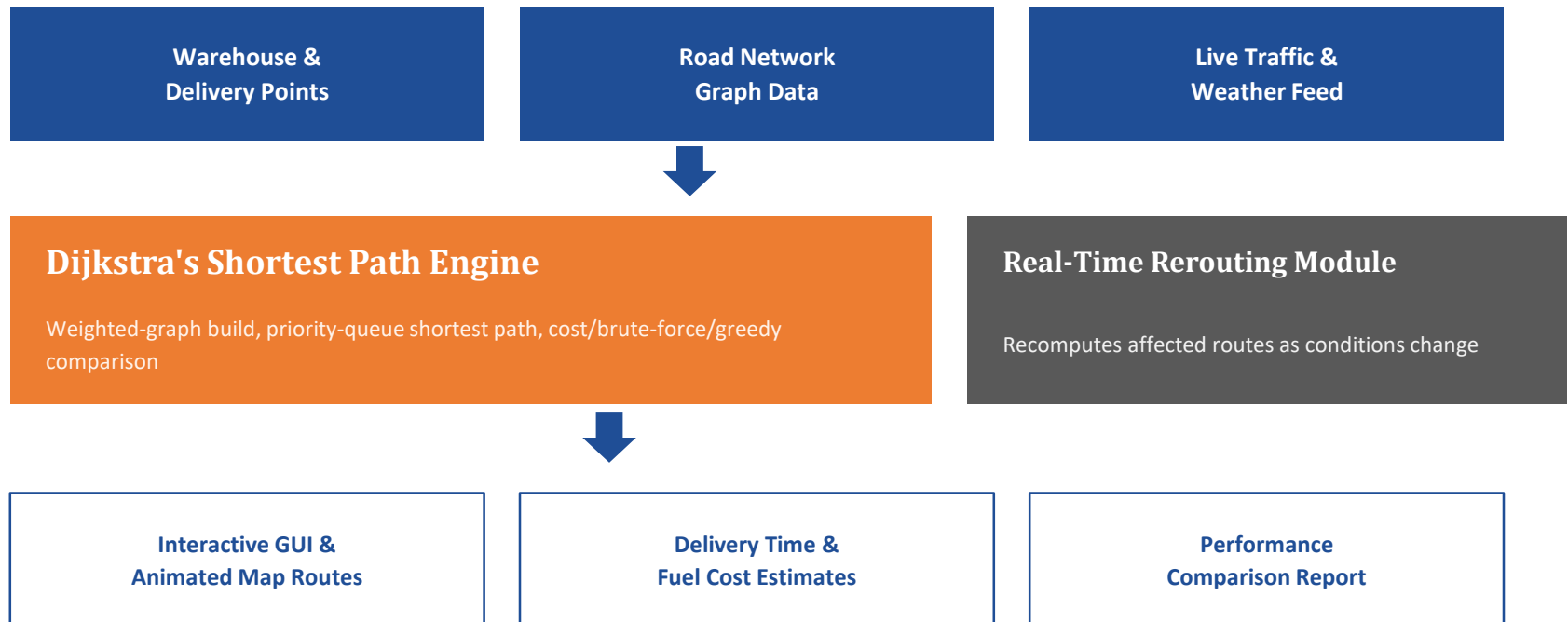


Proposed Objectives

- Design a weighted graph model of cities, roads, and delivery points, with the warehouse as the source node.
- Implement Dijkstra's Algorithm with a priority queue to compute shortest paths to every delivery point in one run.
- Build a GUI for map input with animated visualization of computed routes.
- Estimate delivery time and fuel cost per route, and benchmark against brute-force and greedy approaches.
- Integrate live traffic and weather data to support real-time rerouting.



Architecture





Methodology

1

Data Collection & Graph Modeling

Represent the warehouse, delivery points, and roads as a weighted graph.

2

Shortest Path Computation

Run Dijkstra's Algorithm with a priority queue to find the shortest path to every delivery node.

3

GUI Map Visualization

Render the graph and animate the computed optimal route on an interactive map.

4

Cost & Time Estimation

Convert path distance into estimated delivery time and fuel cost.

5

Comparative Benchmarking

Compare Dijkstra's output with brute-force and greedy algorithms.

6

Dynamic Rerouting

Update edge weights from live traffic/weather data and recompute routes in real time.



SDG

SDG 09

Industry, Innovation & Infrastructure

Applies algorithmic innovation
to modernize logistics
infrastructure.

SDG 11

Sustainable Cities & Communities

Optimized routing reduces
urban traffic congestion.

SDG 12

Responsible Consumption & Production

Shortest-path routing minimizes
fuel usage and waste.

SDG 13

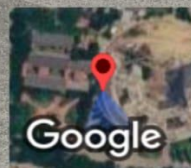
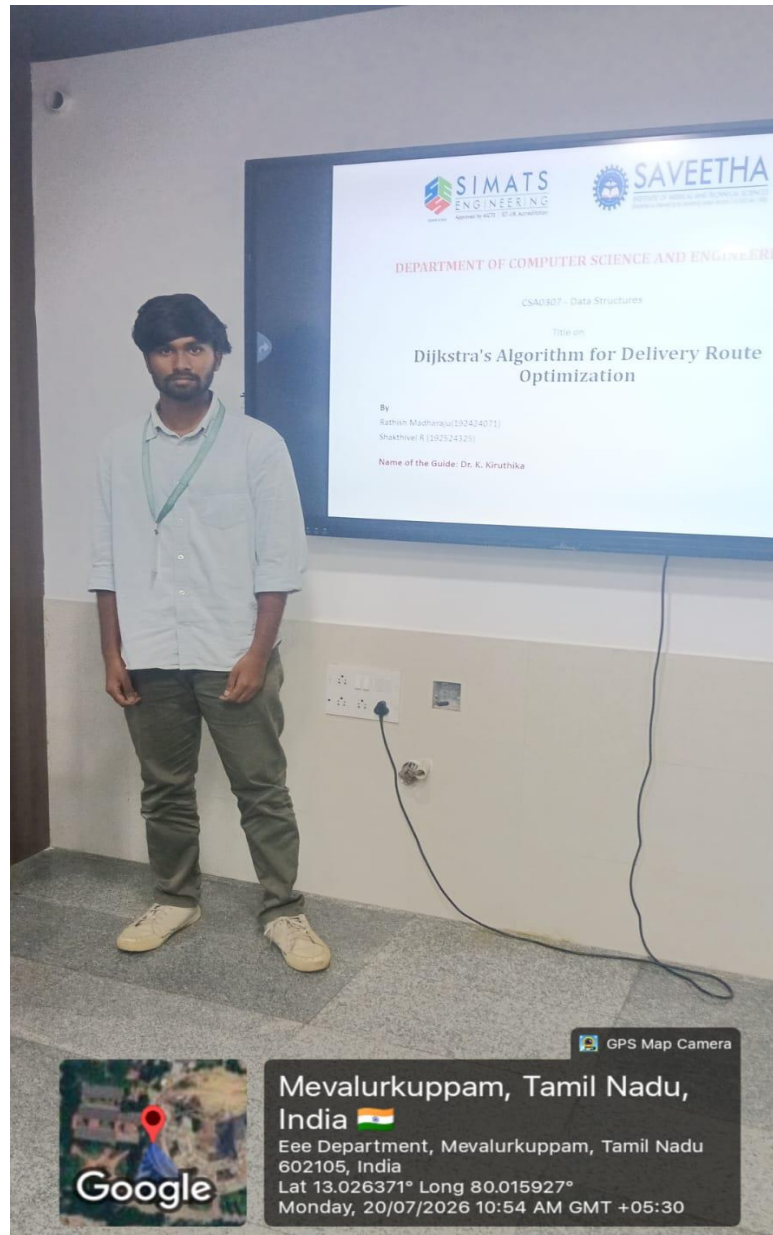
Climate Action

Lower fuel use from optimized
routes reduces carbon
emissions.



Conclusions

- Dijkstra's Algorithm reliably computes the shortest, most cost-efficient delivery paths from a warehouse to multiple destinations.
- The GUI and animated route display make optimized paths easy to interpret for dispatch teams.
- Benchmarking against brute-force and greedy methods confirms Dijkstra's balance of accuracy and efficiency.
- Real-time traffic and weather integration enables adaptive rerouting under changing conditions.
- The approach applies directly to e-commerce, postal services, and food delivery.



GPS Map Camera

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India 🇮🇳

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602105, India
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Monday, 20/07/2026 10:54 AM GMT +05:30